

# SWEEP rule on the KD-Toric code

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## 1 Shrinking

consider changing the name to something like hight lines / contour in order to hint on the similarity to topography map. In addition, the embedding inside  $\mathbb{R}^d$  should appears before that definition.

**Definition 1.1** (Potential Walls). We define the potential walls to be a functions collection from the vertices of  $\mathcal{G}_n^d$  to the reals as follow:

1. For any coordinate  $i \in [d]$ , we define  $L_i: \mathcal{G}_n^d \rightarrow \mathbb{R}$  to act as  $L_i(v) = v_i$ .
2. For any subset of coordinates  $I \subset [d]$  we define  $L_I: \mathcal{G}_n^d \rightarrow \mathbb{R}$  to act as  $L_I(v) = -\sum_{i \in I} v_i$ .

For a subset of vertices  $\Gamma \subset \mathcal{G}_n^d$ , a vertex  $v \in \Gamma$ , and a potential wall  $l: \mathcal{G}_n^d \rightarrow \mathbb{R}$ , we say that  $v$  is a *local maximum* of  $l$  in  $\Gamma$  if for any neighbor  $u$  of  $v$ , such that  $u \in \Gamma$ , we have  $l(v) \geq l(u)$ . We say that  $v$  is a strong local maximum if the inequality is strict, namely  $l(v) > l(u)$ . When the subset  $\Gamma$  is clear from the context, we might omit it and briefly say that  $v$  is a local maximum of  $l$ .

What we want is that if a bit contains both  $v, gv$  then there will be a check that contains both  $v, gv$ , that is true if  $d' > 1$ , then it means that we can erase from the bit a vertex which it is not  $gv$ .

**Claim 1.2.** Let  $z$  be an assignment, and let  $\partial z$  be the set of vertices adjacent to unsatisfied checks induced by  $z$ . Let  $v$  be a vertex in  $\partial z$ , and let  $g$  be a coordinate in  $S$  such that  $v$  is a local maximum of  $L_g$ . Then  $gv \notin \partial z$ .

*Proof.* Suppose  $gv \in \partial z$ , then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to  $v$  being a local maximum of  $L_g$  in  $\partial z$ . □

**Claim 1.3.** Let  $z$  be an assignment, and let  $\partial z$  be the set of vertices adjacent to unsatisfied checks induced by  $z$ . Let  $v$  be a vertex in  $\partial z$ , and let  $g$  be a coordinate in  $S$  such that  $v$  is a local maximum of  $L_g$ . Then for any check associated with a  $d' - 1$  face rooted at  $v$  that contains  $gv$ , it is satisfied.

*Proof.* Assume, through contradiction, the existence of an unsatisfied check associated with a  $d' - 1$  positive face rooted at  $v$ , which contains  $gv$ . This implies  $gv \in \partial z$ , contradicting Claim 1.2.  $\square$

**Definition 1.4** (Small Code at Vertex  $v$ ). For a vertex  $v$ , we define the  $X$ -small code at vertex  $v$  to be the binary assignments over the bits ( $d'$  faces) rooted at  $v$  that satisfy  $\mathcal{C}_X$ 's restrictions rooted at  $v$  ( $d' - 1$  faces). Similarly, we define the  $Z$ -small code at vertex  $v$  to be the binary assignments over the bits ( $d'$  faces) rooted at  $v$  that satisfy  $\mathcal{C}_Z$ 's restrictions rooted at  $v$  ( $d' + 1$  faces). We denote by  $\mathcal{C}_v^-$  and  $\mathcal{C}_v^+$  the  $X$ -small code at  $v$  and  $Z$ -small code at  $v$ .

[Do something with that.](#) And clearly  $\text{Stab}_v$  is collection of stabilizers since for any  $S' \subset S$  such  $|S'| = d' + 1$  we have:

$$\begin{aligned} \partial(\partial\theta_{\mathbf{v}, S'}) &= \partial \left( \sum_{g \in S'} \theta_{\mathbf{v}, S'/g} + \sum_{g \in S'} \theta_{\mathbf{g}\mathbf{v}, S'/g} \right) \\ &= \sum_{\substack{g', g \in S' \\ g' \neq g}} (\theta_{\mathbf{v}, S'/\{g, g'\}} + \theta_{\mathbf{g}'\mathbf{v}, S'/\{g, g'\}} + \theta_{\mathbf{g}\mathbf{v}, S'/\{g, g'\}} + \theta_{\mathbf{g}'\mathbf{g}\mathbf{v}, S'/\{g, g'\}}) \\ &= 0 \end{aligned}$$

Where we used that the term in the closure is invariant to sweeping between  $g$  and  $g'$ , and therefore any term in the sum appears twice.

**Definition 1.5.** [Should rewrite it better.](#) We define  $\text{Stab}_v^-$  and  $\text{Stab}_v^+$  as the following linear codes:

$$\begin{aligned} \text{Stab}_v^- &:= \left\{ z : \partial z = 0 \text{ and } \right. \\ &\quad \left. z \in \text{Span} \left( \left\{ \theta_{\mathbf{v}, S'/g}, \theta_{\mathbf{g}\mathbf{v}, S'/g} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \right\} \right) \right\} \\ \text{Stab}_v^+ &:= \left\{ z : \partial^* z = 0 \text{ and } \right. \\ &\quad \left. z \in \text{Span} \left( \left\{ \theta_{\mathbf{v}, S'/g}, \theta_{\mathbf{g}\mathbf{v}, S'/g} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \right\} \right) \right\} \end{aligned}$$

**Claim 1.6.** Let  $v$  be a vertex. For any  $z \in \mathcal{C}_v^\pm$ , there is a stabilizer  $w \in \text{Stab}_v^\pm$  such that  $z = w|_v$ ; namely,  $z$  and  $w$  agree on the faces rooted at  $v$ .

*Proof.* We will show that codes obtained from  $\text{Stab}_v^-$  and  $\text{Stab}_v^+$  by forcing the faces rooted at  $v$  to  $z|_v$  have non-zero dimensions. Then it follows that any code word of  $\mathcal{C}_v^\pm$  could be extend to codewords in  $\text{Stab}_v^-$  and  $\text{Stab}_v^+$ . The number of bits in both  $\text{Stab}_v^\pm$  is:

$$\overbrace{\binom{d}{d'}}^{\text{bits of the form } \theta_{\mathbf{v}, S'/g}} + \overbrace{d \binom{d-1}{d'}}^{\text{bits of the form } \theta_{\mathbf{g}\mathbf{v}, S'/g}}$$

The numbers of checks for  $\text{Stab}_\pm$  are:

$$\overbrace{\binom{d}{d' \pm 1}}^{\text{checks of the form}} + \overbrace{d \binom{d-1}{d' \pm 1}}^{\text{checks of the form}}$$

The restriction to  $z|_v$  sets the bits rooted at  $v$ , and satisfies the checks that are rooted at  $v$ , so the checks of the form  $\theta_{\mathbf{v}, \mathbf{S}'}$  for  $|S'| = d' \pm 1$  are degenerate, and the dimension that remains is:

$$\dim \geq d \binom{d-1}{d'} - d \binom{d-1}{d' \pm 1}$$

In the case that  $d' = d/2$ , then for any  $x \in [d-1]$  we have  $\binom{d-1}{d'} > \binom{d-1}{x}$  therefore:

$$\begin{aligned} \dim &\geq d \binom{d-1}{d/2} - d \binom{d-1}{d/2 \pm 1} \\ &\geq d \end{aligned}$$

So there is at least one stabilizer  $w \in \text{Stab}_v$  that agrees with  $z$ .  $\square$

**Claim 1.7.** Let  $z$  be a finite assignment. Suppose that  $\max_{v \in z} L_{d+1}(v) > \max_{v \in \partial z} L_{d+1}(v)$ . Then there is  $\tilde{z} \sim_{\mathcal{C}_X} z$  such that  $\max_{v \in \tilde{z}} L_{d+1} < \max_{v \in z} L_{d+1} - 1$ .

*Proof.* Let  $v_1, v_2, \dots, v_k$  be the vertices in  $z$  that achieve the maximum for  $L_{d+1}$ , namely  $v_1, v_2, \dots, v_k = \arg \max_{v \in z} L_{d+1}(v)$ , and assume that  $v_i \notin \partial z$  for any  $i \in [k]$ . Consider the  $i$ th vertex  $v_i$ , by definition, since  $v_i \notin \partial z$ , we have  $(\partial z|_{v_i})|_{v_i} = 0$ , and since  $v_i$  is a maximum point of  $L_{d+1}$  in  $z$ , it is also a local maximum, and therefore  $z|_{v_i+} = z|_{v_i}$ . Combining the two, we get that  $z|_{v_i}$  is a codeword of the small code at  $v$ , namely  $z|_{v_i} \in \mathcal{C}_{v_i}$ . Thus by Claim 1.6, there is a stabilizer  $w_i \in \text{Stab}_{v_i}$  such that  $w_i$  and  $z$  agree on the positive faces of  $v_i$ , namely  $w_i|_{v_i} = z|_{v_i}$ . Set  $\tilde{z} \leftarrow z + \sum_i w_i$ . Since  $\sum_i w_i$  is a stabilizer, we have  $\tilde{z} \sim_{\mathcal{C}_X} z$ . In addition, since for any  $j \neq i$ ,  $v_j \notin w_i$  we have  $\tilde{z}|_{v_i} = (z + w_i)|_{v_i} = 0$  namely  $v_i \notin \tilde{z}$ , thus:

$$\max_{u \in \tilde{z}} L_{d+1}(u) \leq \max_{u \in z \cup \{w_i\} \cup v_i} L_{d+1}(u) < \max_{u \in z} L_{d+1}(u) - 1$$

$\square$

**Claim 1.8.** Let  $z$  be a finite assignment. Then there is  $\tilde{z} \sim_{\mathcal{C}_X} z$  such that  $\max_{v \in \tilde{z}} L_{d+1} = \max_{v \in \partial \tilde{z}} L_{d+1}$ .

*Proof.* Apply Claim 1.7 multiple times.  $\square$

**Claim 1.9.** Let  $z$  and  $\xi$  be the assignment and the result of the application of the Sweep rule on it. Then:

1. For any  $i \in [d]$ ,  $\max_{v \in \partial z} L_i(v) = \max_{v \in \partial \xi} L_i(v)$ .

$$2. \max_{v \in \partial z} L_{d+1}(v) > \max_{v \in \partial \xi} L_{d+1}(v) + 1.$$

*Proof.* First, since the Sweep rule is invariant to stabilizers, for any  $\tilde{z}$  such that  $\tilde{z} \sim_{C_X} z$ , the correction applied by the Sweep rule on  $\tilde{z}$  is the same as the application on  $z$ . Thus,  $\partial \tilde{\xi} = \partial \xi$ , and we can assume by Claim 1.8 that  $\max_{v \in z} L_{d+1} = \max_{v \in \partial z} L_{d+1}$ .

Thus, the local view around the edge  $v$  of  $\partial z$  is rooted at  $v$ , therefore,  $v$  can flip by that and local view and get  $v \notin z \rightarrow v \notin \partial z$ , and eventually we repeat on the same max inequality above to show that  $L_{d+1}$  decreases.  $\square$

*Proof.*

$$L_{d+1}(u) \leq \min_{v \in \partial z} L_{d+1}(v)$$

Observes that if the *termination-condition* holds then any local maximum of  $L_{d+1}$  in  $\partial z$  is also a local maximum of  $L_{d+1}$  in  $\tilde{z}$  and we finish. Consider the other case, namely there is at least one vertex  $u \in \tilde{z}/\partial z$  such that  $u$  is a local maximum for  $L_{d+1}$  and

$$L_{d+1}(u) > \min_{v \in \partial z} L_{d+1}(v)$$

In that case, we peak such  $u$ , and looking for stabilizer  $w$  such  $u \notin \tilde{z} + w$  by repeating over the above procedure where  $u$  plays the role of  $v$ . We keep repeating on the procedure until the *termination-condition* holds.

We argue that the process indeed terminates. Consider a vertex  $u$  such that  $u$  is a local maximum of  $L_{d+1}$  in  $\tilde{z}$  and  $u$  is not a local maximum of  $L_{d+1}$  at  $z$ . Put differently,  $u$  becomes a local maximum as a result of adding  $w$  to  $z$ . Thus,  $u$  has to belong to  $w|_v \setminus v$ , which implies that there is a non-empty subset of generators  $I \subset S$  such that  $u = (\prod_{g \in I} g)v$ , and therefore  $L_{d+1}(u) \leq -1 + L_{d+1}(v)$ .

Hence, since  $z$  is finite.

We repeat the process while possible; namely, we stop when all the local maxima of  $L_{d+1}$  are in  $\partial$ .  $\square$

We prove that  $\dim \mathcal{C}_v = \dim \text{Stab}_v$  below, for now assume it is correct. [We continue by using the fact that if, for a local maximum  \$v\$  in  \$\partial\$ , it holds that  \$z|\_v\$  is rooted at  \$v\$ , then the decoder will find bits rooted at  \$v\$  such that their  \$\partial\$  agrees with  \$\partial\$ .](#)

To generalize, one might think that it has to complete the local view on  $v$  such that all the checks are satisfied. For  $C_X$ , it sums up to show that the following dimension is positive. Yet the trick fails for  $C_Z$ , where we need to think of a different idea.

$$d \binom{d-1}{d'} - \binom{d}{d'-1} - d \binom{d-1}{d'-1}$$

**Claim 1.10.** Let  $z$  an assignment rooted at vertex  $v$ , such  $\partial z|_{v+} = 0$ . Then there is a stabilizer  $w$ , rooted at  $v$ , such that  $w|_{v+} = z$ .

*Proof.* Denote by  $F$  the faces in  $z$ , and by  $J$  the collection of generators that support  $F$ . We claim that  $|J| \geq d' + 1$ .

Clearly,  $|J| \geq d'$ , since each face is defined by a root vertex and  $d'$  generators. Yet, observe that if  $|J|$  equals exactly  $d'$ , then  $z$  contains a single  $d'$ -face, denote it by  $f$ . Then, any check associated with a  $d' - 1$  face of  $f$  is unsatisfied, in particular the  $d' - 1$  faces that are rooted at  $v$ , in contradiction to  $\partial z|_{v+} = 0$ .

Now, one can choose  $J' \subset J$  of size  $d' + 1$ , and define the stabilizer  $w = \theta_{v,J'}$ .  $\square$

**Claim 1.11.** Suppose  $v \in z$  and for any  $g \in S$ , we have  $g^{-1}v \notin z$ , then there is  $\theta_{v,J} \in \partial z$ .

*Proof.*

$$\partial z = \sum_{S'} \theta_{\mathbf{v}, S'} + ..$$

□

**Claim 1.12.** Let  $z$  be a connected assignment on the  $d'$ -faces at finite size, and let  $\Gamma$  be the set of vertices which are adjacent to unsatisfied checks induced by  $z$ . Let  $v$  be a vertex in  $\mathcal{G}_\infty^d$  such that  $v$  is a strong global maximum of  $L_{d+1}$ , at  $\Gamma$ . Then there is a rooted assignment  $\tilde{z}$  at  $v$  such that  $z_v = \tilde{z}_v$ .

*Proof.* Let  $\Theta = \theta_{v_1, S'_1}, \theta_{v_2, S'_2}, .. \theta_{v_k, S'_k}$  be the faces forms  $z$ .

Since  $z$  is finite, it has a maximum for  $L_{d+1}$ . Denote by  $u$  the vertex that gets the maximum, and denote by  $\theta_{u, J_1}, \dots, \theta_{u, J_l}$  the  $d'$ -faces in  $z$  that are rooted at  $u$ . We claim that  $u = v$ .

Suppose not, namely  $u \notin \Gamma$ . Then it means that any  $d' - 1$  face containing  $u$  is satisfied. Observe that  $\sum \theta_{u, J_i} \neq 0$ ; otherwise, it would contradict the non linearity independence of  $\theta_{v_1, S'_1}, \theta_{v_2, S'_2}, .. \theta_{v_k, S'_k}$ . So, by Claim 1.11 there exists a  $d' - 1$  face rooted at  $u$ , contained in  $\sum \theta_{u, J_i}$ . Let us denote it by  $\theta_{u, J'}$ .

Since  $\theta_{u, J'}$  is satisfied, it has to be adjacent to a turned bit, namely a  $d'$  face in  $z$ , not in  $\theta_{u, J_1}, \dots, \theta_{u, J_l}$ . Denote that  $d'$  face by  $f$ . Yet, since  $\theta_{u, J'}$  is rooted at  $u$ , it contains  $u$ , and therefore  $f$  contains  $u$ . Now, if  $f$  is rooted at  $u$ , then it is a contradiction that  $\theta_{u, J_1}, \dots, \theta_{u, J_l}$  are all the  $d'$  faces in  $z$  rooted at  $u$ . If  $f$  is not rooted at  $u$ , denote its root by  $w$ , so there is  $g \in S$  such that  $gw = u$ , and  $w \in z$ , in contradiction to  $u$  being maximum for  $L_{d+1}$ .

So, we find that  $v$  is also the maximum of  $L_{d+1}$  in  $z$ . Therefore, we have that  $\tilde{z} = \sum \theta_{u, J_i}$  zeros out  $z_v$ . □

**Claim 1.13.** Let  $z$  be a connected assignment on the  $d'$ -faces at finite size, and let the collection  $\Theta = \theta_{v_1, S'_1}, \theta_{v_2, S'_2}, \dots, \theta_{v_k, S'_k}$  be the decomposition forms  $z$ . Denote by  $\Gamma$  the set of vertices adjacent to unsatisfied check induced by  $z$ . Then  $\Theta$  is acyclic, if and only if  $z$  has at least one Infimum.

*Proof.* First, suppose that  $z$  has at least one Infimum. Let  $u$  be an infimum in  $z$ . Second, suppose that  $\Theta$  is acyclic.  $\square$

**Claim 1.14.** Suppose that  $d' > 1$  and a vertex  $v \in \mathcal{G}_\infty^d$  is also adjacent to the domain wall, namely  $v \in \Gamma$ . It holds that  $v$  cannot be simultaneously both a (strong) local max of  $L_{d+1}$  and a local min of  $L_{d+1}$ .

*Proof.* Let  $v \in \Gamma$  be a local maximum point of  $L_{d+1}$ , It means that:

1. The vertex  $v$  is adjacent to a  $d'^{-1}$ -face which is unsatisfied.
2. That face is rooted in  $v$ ; otherwise, the root of the face is a vertex in  $\Gamma$  with a higher value of  $L_{d+1}$ , in contradiction to  $v$  being a local maximum.

Thus, there is a subset of generators  $S' \subset S$ , of size  $d' - 1 > 0$ , such that the unsatisfied check set is on the face  $\theta_{v, S'}$ . Given  $d' > 1$ , we have that  $S'$  contains at least one more element besides the empty set, denoted by  $g$ . Now, consider the vertex  $gv$ . Since  $gv$  is adjacent to  $\theta_{v, S'}$ , then  $gv \in \Gamma$ . On the other hand, on the infinite toric  $\mathcal{G}_\infty^d$ , moving along the grid generator decreases  $L_{d+1}$  by one, so  $L_{d+1}(gv) = L_{d+1}(v) - 1$ . In other words,  $gv$  is a neighbor of  $v$  with a lower value of  $L_{d+1}$ , and thus  $v$  cannot be a local minimum of  $L_{d+1}$ .  $\square$

**Remark 1.15.** That claim alone gives a decoder in the clean setting. Write a remark indicating that when  $d' = 1$ , meaning qubits are on the edges (2D toric), the claim is not true.

For  $d = 4, d' = 2$ . For  $i, j \in [d]$ , such that  $i \neq j$ , denote the bit rooted at  $v$  and spanned by generators  $g_i, g_j \in S$ , namely  $\theta_{v, \{g_i, g_j\}}$ , by  $g_{ij} = g_{ji}$ . The restrictions:

$$r_i: \sum_{j \neq i} g_{ij}$$

Thus we get

$$\begin{aligned} \sum_i^n r_i &= \sum_i^n \sum_{j \neq i}^{n+1} g_{ij} \\ &= \sum_i^n \sum_{j < i}^n g_{i,j} + g_{j,i} + \sum_i^n g_{i,n+1} \\ &= r_{n+1} \end{aligned}$$

Thus the dimension of the small code at  $v$  is at least:

$$\binom{d}{2} - (d - 1) = 3$$

We have  $\binom{4}{3} = 4$  stabilizers rooted at  $v$  (each for each cube). Yet:

$$s_1 = g_{1,2} + g_{2,3} + g_{1,3}$$

$$s_2 = g_{1,2} + g_{2,4} + g_{1,4}$$

$$s_3 = g_{1,3} + g_{3,4} + g_{1,4}$$

$$s_4 = g_{2,3} + g_{3,4} + g_{2,4}$$

And clearly

$$s_4 = s_1 + s_2 + s_3$$

**Definition 1.16** (Reduced subset with boundrey at  $v$ ).